

Problem Statement

International airport operations have crucial needs in order to coordinate across multiple independent enterprises, each with unique responsibilities and employees. If an integrated platform is missing, communication between airlines, airport authorities, and ground service providers is fragmented. This leads to delayed gate assignments, missed ground handling windows, and passenger disruptions that are unresolved.

This project offers a digital ecosystem platform for JFK International Airport that connects three enterprises, Japan Airlines, New York Port Authority, and GroundMaster services. The ecosystem has been carefully designed and built on a unified, role-based communication and work request system. This platform enables cross-enterprise coordination that no single enterprise can achieve alone and delivers value greater than the sum of its parts.

These enterprises are dependent on each other because Japan Airlines has the aircraft and passengers, but no airport infrastructure and ground workers. The New York Port Authority owns and operates the infrastructure but has no airlines to utilize it. GroundMaster services has the workforce but no direction without flight schedules and gate assignments. No matter the flight, every departure depends on all three of these entities working in smooth coordination. One missed handoff or conflict turns into delays, missed connections, and passenger disruptions.

Team Contributions

Person 1 (Fred Triest)

- Create class hierarchy, role based authentication, System Admin and Enterprise Admin CRUD panels, ConfigureASystem with pre-populated test data, high level diagram, project proposal.

Person 2 (Cohen Powell)

- Work request engine with all 6 request types and status transitions, cross-enterprise and cross-organization routing, work area UI panels for all 8 operational roles. Presentation slides.

Person 3 (Larry Tsao)

- Faker data integration, reporting module with analytics and summaries, UML diagram.

The Enterprises

Japan Airlines

The main carrier operating out of JFK. JAL manages flight schedules, passenger services, and coordinates with the port authority on gate assignments and operational permits. JAL cannot function without the infrastructure and ground services the other enterprises offers.

New York Port Authority

The governing body that owns and operates JFK airport. This enterprise allocates gates, issues with operational permits, manages the capacity of terminals, and enforces safety and compliance standards. No airplane takes off without the port authority's authorization. The port authority is the central coordinator of the entire platform.

GroundMaster Services

The private company that is a contractor is responsible for servicing the airplanes between flights. This enterprise refuels planes, loads baggage, cleans cabins, and operates all ramp operations. Every aircraft turnaround depends on GroundMaster. If this enterprise was not here, no aircraft is ready for its next departure regardless of how well the gate is assigned or the permit is issued.

Organizations

Enterprise	Organization	Description
Japan Airlines	Flight Operations Division	Manages flight scheduling and operational activities
Japan Airlines	Passenger Services Department	Manages check-in, boarding, and passenger complaints
New York Port Authority	Gate & Terminal Office	Allocates gates and manages terminal capacity
New York Port Authority	Safe & Compliance Bureau	Performs safety inspections and issues permits
GroundMaster Services	Ramp & Refueling Unit	Manages ramp crews and aircraft refueling

GroundMaster Services	Baggage & Cabin Services Unit	Handles baggage loading and cabin turnaround
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Roles

Role	Enterprise	Organization
Flight Scheduler	Japan Airlines	Flight Operations Division
Operations Manager	Japan Airlines	Flight Operations Division
Passenger Services Agent	Japan Airlines	Passenger Services Dept
Gate Coordinator	New York Port Authority	Gate & Terminal Office
Safety Inspector	New York Port Authority	Safety & Compliance Bureau
Compliance Officer	New York Port Authority	Safety & Compliance Bureau
Ramp Supervisor	GroundMaster Services	Ramp & Refueling Unit
Baggage Handler	GroundMaster Services	Baggage & Cabin Services Unit
System Admin	Platform	EcoSystem Level
Enterprise Admin	Platform	Enterprise Level

Work Requests

Request Type	Sender	Receiver	Scope	Statuses
Gate Assignment Request	Flight Scheduler	Gate Coordinator	Cross-enterprise	Submitted > Under Review > Assigned/Conflict Flagged > Confirmed
Ground	Flight Scheduler	Ramp	Cross-enterprise	Dispatched >

Handling Request		Supervisor		Crew Assigned > In Progress > Completed
Operational Permit Request	Operations Manager	Safety Inspector	Cross-enterprise	Submitted > Under Review > Approved/Conditionally Approved/Denied
Baggage Irregularity Report	Baggage Handler	Ramp Supervisor	Cross-organization (GroundMaster)	Filed > Under Investigation > Escalated/Resolved
Incident Report and Safety Audit	Safety Inspector	Compliance Officer	Cross-organization (NYPA)	Reported > Under Review > Audit Initiated/Closed/Violation Issued
Passenger Disruption Request	Passenger Services Agent	Operations Manager	Cross-organization (JAL)	Escalated > Under Review > Authorized/Denied > Resolved

Key Architectural Decisions

- Singleton Pattern - Airport EcoSystem is implemented as a Singleton ensuring only one instance of the ecosystem exists at runtime, preventing duplicate data and authentication conflicts.
- Abstract Class Hierarchy - Role and Organization are abstract classes, enforcing that every subclass implements createWorkArea() and getSupportedRole() respectively.
- Role-based Authentication - login searches user accounts across three levels of the hierarchy (ecosystem, enterprise, organization). Passwords are never exposed through a getter. validateUser() method returns boolean to confirm validation.
- CardLayout Navigation - work area panels are managed with a single JFrame using CardLayout, which provides a single window experience with named card routing based on the user's role.
- Encapsulation and Validation - All fields are private with validated setters. Email addresses are validated against a regular expression pattern. Age is validated against legal employment bounds of 16 to 100.

- Unique ID Generation - every Employee, UserAccount, Organization, Enterprise, and Network receives an auto-generated ID using a static counter shared across all instances of each class.

Test Users

Username	Password	Role
sysadmin	sysAdmin1&	System Admin
jaladmin	jalAdmin1&	Enterprise Admin (Japan Airlines)
nypaadmin	nypaAdmin1&	Enterprise Admin (NY Port Authority)
groundadmin	groundAdmin1&	Enterprise Admin (GroundMaster)
flightScheduler	flightScheduler1&	Flight Scheduler
operationsManager	operationsManager1&	Operations Manager
passengerAgent	passengerAgent1&	Passenger Services Agent
gateCoordinator	gateCoordinator1&	Gate Coordinator
safetyInspector	safetyInspector1&	Safety Inspector
complianceOfficer	complianceOfficer1&	Compliance Officer
rampSupervisor	rampSupervisor1&	Ramp Supervisor
baggageHandler	baggageHandler1&	Baggage Handler